

Data Handling and Visualization

Class Activity

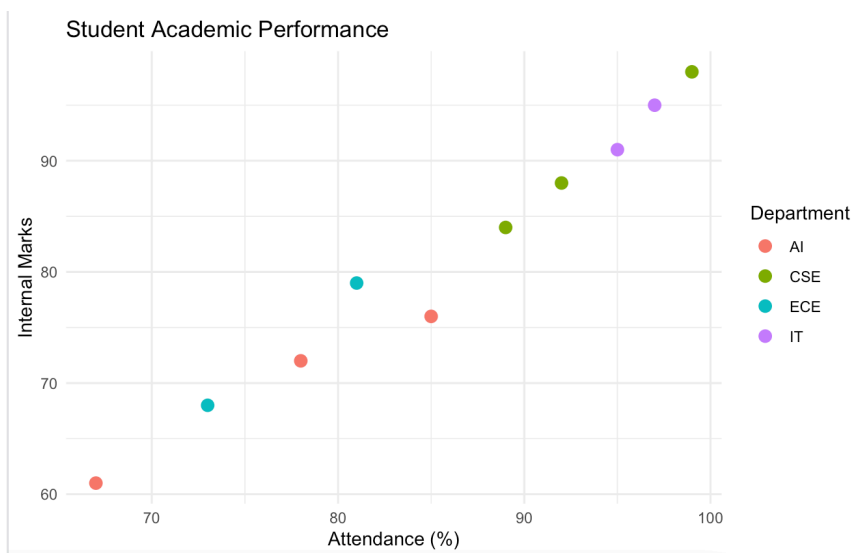
Date: 13/07/2026

Scenario 1:

```
student <- read.csv("downloads/data/student_data.csv")

ggplot(student, aes(x = Attendance, y = Internal_Marks, color = Department)) +
  geom_point(size = 3) +
  labs(
    title = "Student Academic Performance",
    x = "Attendance (%)",
    y = "Internal Marks",
    color = "Department"
  ) +
  theme_minimal()
```

Output:

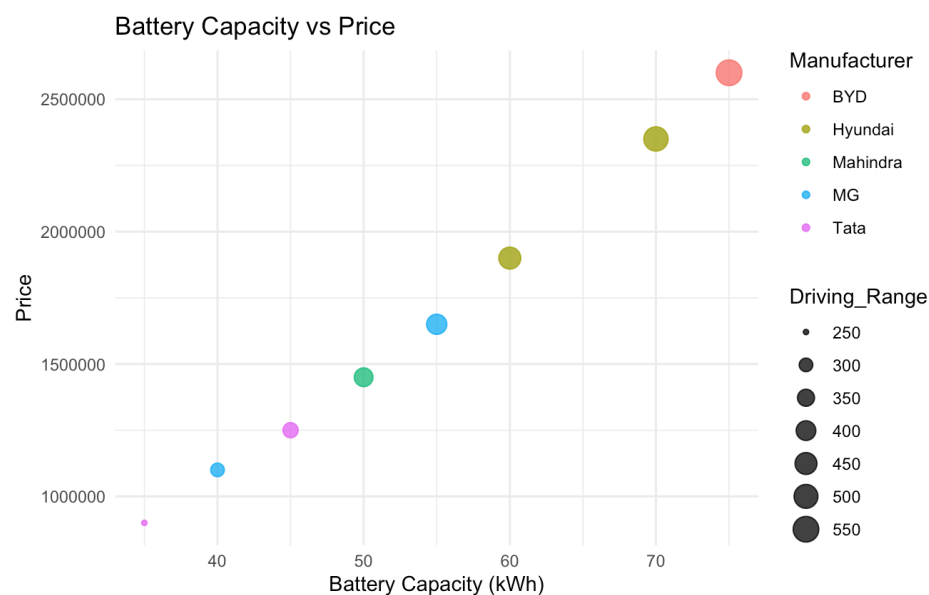


Scenario 2:

```
ev <- read.csv("downloads/data/ev_data.csv")

ggplot(ev, aes(x = Battery_Capacity, y = Price,
               size = Driving_Range,
               color = Manufacturer)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Battery Capacity vs Price",
    x = "Battery Capacity (kWh)",
    y = "Price"
  ) +
  theme_minimal()
```

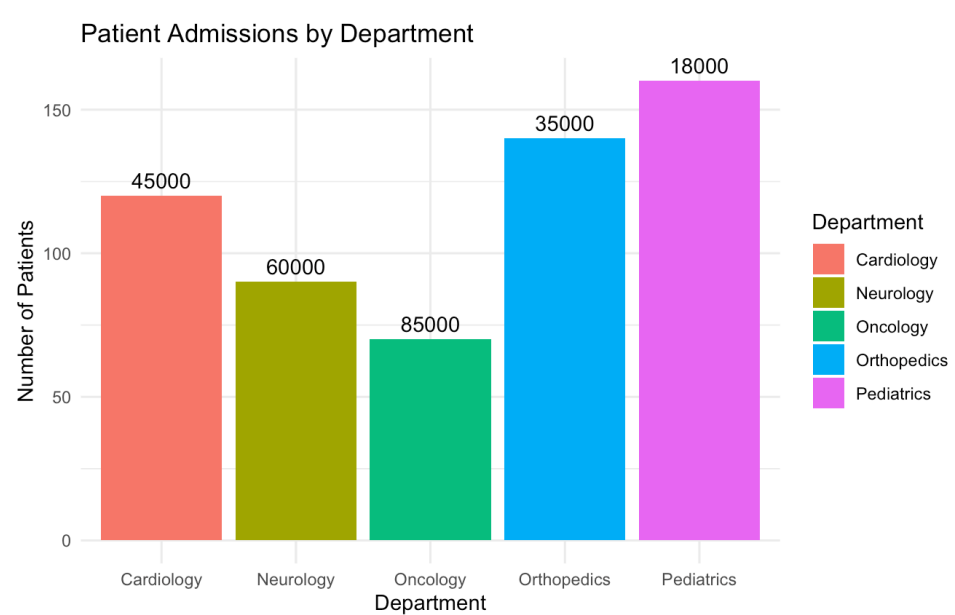
Output



Scenario 3:

```
hospital <- read.csv("downloads/data/hospital_data.csv")
ggplot(hospital,
  aes(x = Department,
    y = Number_of_Patients,
    fill = Department)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = Average_Treatment_Cost),
    vjust = -0.5) +
  labs(
    title = "Patient Admissions by Department",
    x = "Department",
    y = "Number of Patients"
  ) +
  theme_minimal()
```

Output

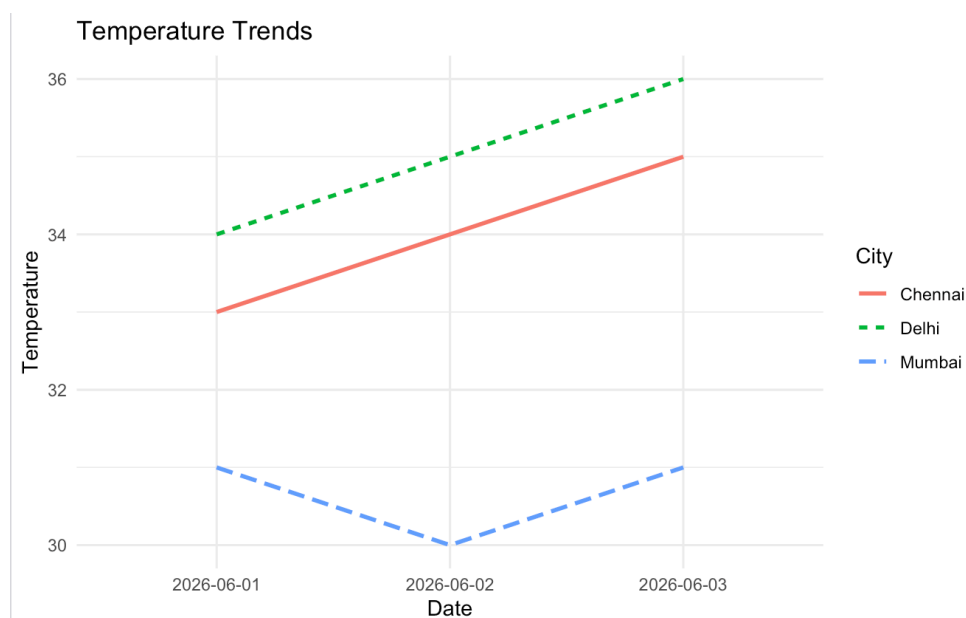


Scenario 4:

```
weather <- read.csv("downloads/data/weather_data.csv")
```

```
ggplot(weather,  
  aes(x = Date,  
      y = Temperature,  
      color = City,  
      linetype = City,  
      group = City)) +  
  geom_line(size = 1) +  
  labs(  
    title = "Temperature Trends",  
    x = "Date",  
    y = "Temperature"  
  ) +  
  theme_minimal()
```

Output

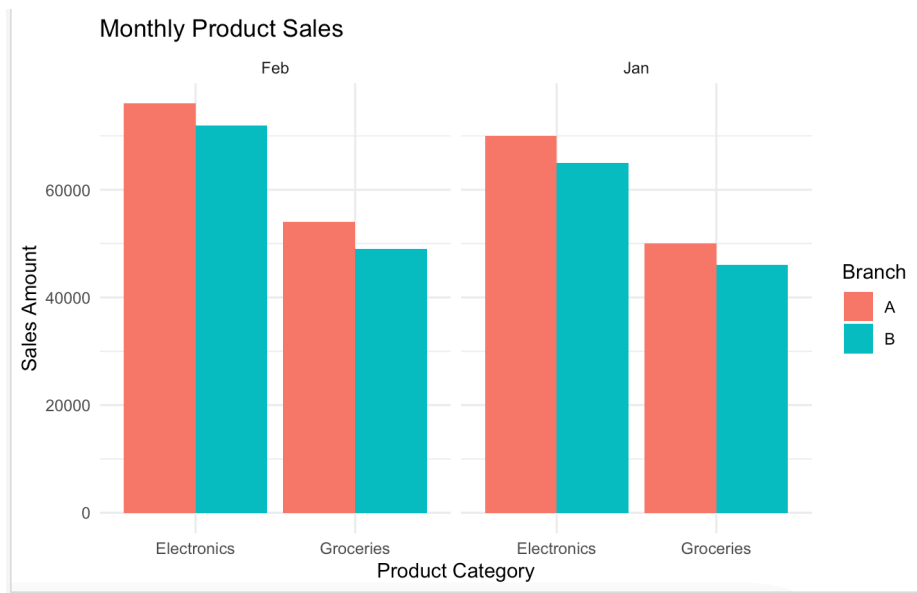


Scenario 5:

```
sales <- read.csv("downloads/data/sales_data.csv")
```

```
ggplot(sales,  
  aes(x = Product_Category,  
      y = Sales_Amount,  
      fill = Branch)) +  
  geom_bar(stat = "identity",  
    position = "dodge") +  
  facet_wrap(~Month) +  
  labs(  
    title = "Monthly Product Sales",  
    x = "Product Category",  
    y = "Sales Amount"  
  ) +  
  theme_minimal()
```

Output

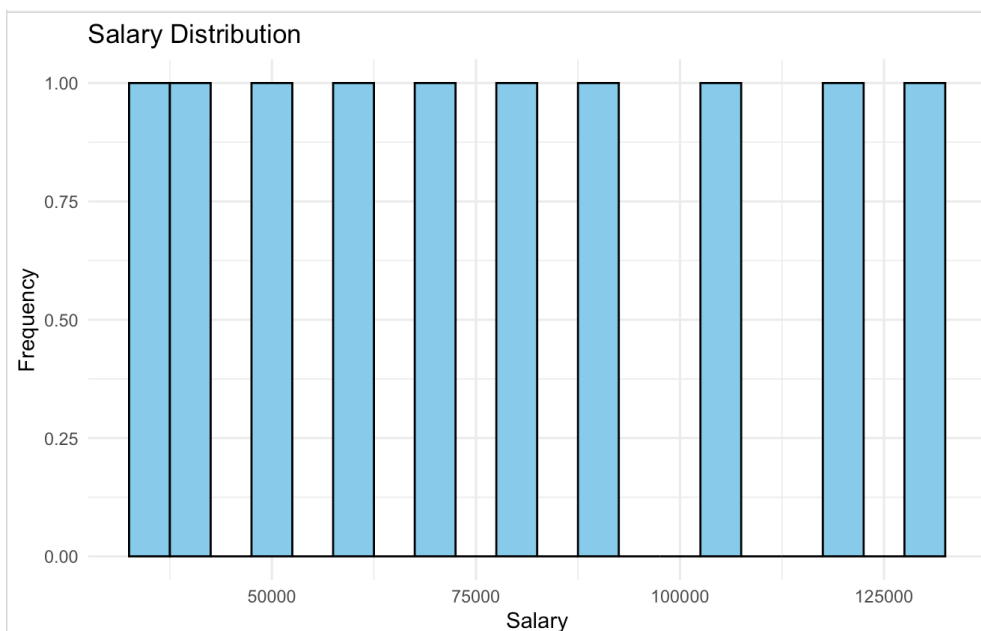


Scenario 6:

```
employee <- read.csv("downloads/data/employee_data.csv")
```

```
ggplot(employee,  
  aes(x = Salary)) +  
  geom_histogram(binwidth = 5000,  
    fill = "skyblue",  
    color = "black") +  
  labs(  
    title = "Salary Distribution",  
    x = "Salary",  
    y = "Frequency"  
  ) +  
  theme_minimal()
```

Output

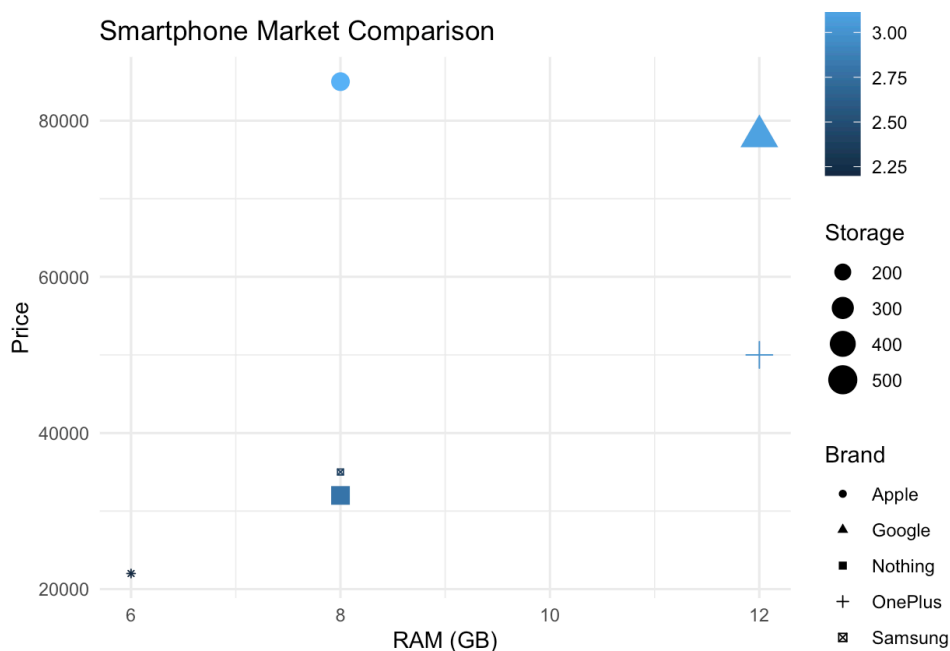


Scenario 7:

```
phone <- read.csv("downloads/data/smartphone_data.csv")
```

```
ggplot(phone,  
  aes(x = RAM,  
      y = Price,  
      shape = Brand,  
      size = Storage,  
      color = Processor_Speed)) +  
geom_point() +  
labs(  
  title = "Smartphone Market Comparison",  
  x = "RAM (GB)",  
  y = "Price"  
) +  
theme_minimal()
```

Output



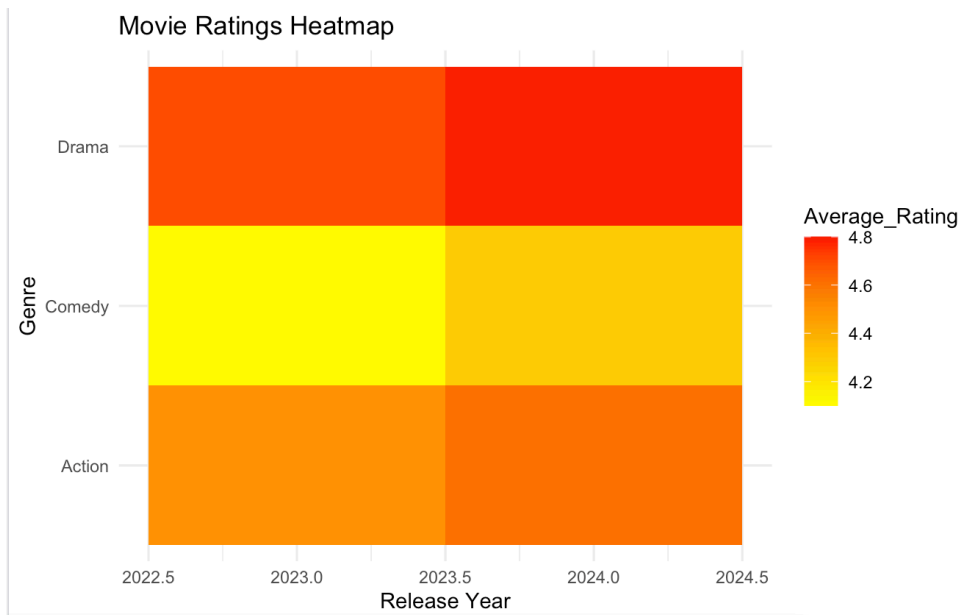
Scenario 8:

```
movie <- read.csv("downloads/data/movie_data.csv")
```

```
ggplot(movie,  
  aes(x = Release_Year,  
      y = Genre,  
      fill = Average_Rating)) +  
geom_tile() +  
scale_fill_gradient(low = "yellow",  
  high = "red") +  
labs(  
  title = "Movie Ratings Heatmap",  
  x = "Release Year",  
  y = "Genre"  
) +
```

```
theme_minimal()
```

Output

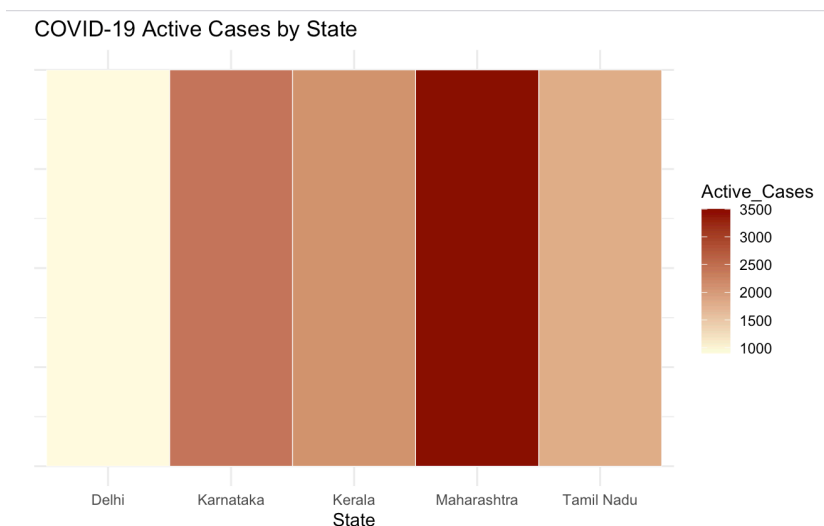


Scenario 9:

```
covid <- read.csv("downloads/data/covid_data.csv")
```

```
ggplot(covid,  
  aes(x = State,  
      y = 1,  
      fill = Active_Cases)) +  
  geom_tile(color = "white") +  
  scale_fill_gradient(low = "lightyellow",  
                     high = "darkred") +  
  labs(  
    title = "COVID-19 Active Cases by State",  
    x = "State",  
    y = ""  
  ) +  
  theme_minimal() +  
  theme(axis.text.y = element_blank(),  
        axis.ticks.y = element_blank())
```

Output



Scenario 10:

```
ggplot(iris,
  aes(x = Sepal.Length,
      y = Petal.Length,
      color = Species,
      shape = Species,
      size = Sepal.Width)) +
  geom_point() +
  labs(
    title = "Iris Flower Classification",
    x = "Sepal Length",
    y = "Petal Length"
  ) +
  theme_minimal()
```

Output

